

Foundations

§1

Stability certificates

§2

Dynamical system (X, T, φ_t)

phase space X ; time set $T \subseteq \mathbb{R}$ or $\mathbb{Z}$; evolution operators $\varphi_t : X \rightarrow X$ with **(D1)** $\varphi_0 = \text{Id}$ and **(D2)** $\varphi_t \circ \varphi_s = \varphi_{t+s}$

Kinds of time

discrete: $T \cong \mathbb{N}_0, \mathbb{Z}$ — iterated maps g^k ; **continuous:** T an interval — flows; **invertible:** T a group, $\varphi_{-t} = \varphi_t^{-1}$.
logistic map $px(1-x)$: inverse is two-valued $\Rightarrow$ not invertible

Orbits & invariant sets

Or(x_0) = $\{\varphi_t(x_0) : t \in T\}$; periodic: $\varphi_{t+T_0}(x_0) = \varphi_t(x_0)$; the **phase portrait** is the decomposition of X into orbits.
 S **invariant:** $\varphi_t(x_0) \in S$ for all t , all $x_0 \in S$ (**positively/negatively:** $t > 0$ / $t < 0$).
nullclines of $x' = f(x)$: $\{x : f_k(x) = 0\}$ — organize the phase portrait. van der Pol: $x' = y - \frac{x^3}{3} + x, y' = -px$: cubic + vertical line

Existence & uniqueness

f smooth $\Rightarrow x' = f(x), x(0) = x_0$ has a unique smooth solution on a maximal interval $(-t_1, t_2)$ — via Banach's fixed-point theorem (contraction on a complete metric space).
maximal $\neq$ global: $x' = x^2$ blows up, $\varphi_t(x_0) = \frac{1}{1/x_0 - t}$ — only a *local* flow

Stability of equilibria

§2

0 Equilibrium (fixed point)

$\varphi_t(x^*) = x^* \quad \forall t \in T$
ODE: $f(x^*) = 0$; map: $g(x^*) = x^*$

1 Lyapunov stable

$\forall U \exists V \subseteq U : \varphi_t(V) \subseteq U \quad \forall t \geq 0$
nearby orbits stay nearby

2 Attracts a positive-measure set

$\mu(B(x^*)) > 0$
basin $B(A) := \{x_0 : \omega(x_0) \subset A\}$

3 Attracts a whole neighborhood

$\varphi_t(x_0) \xrightarrow{t \rightarrow +\infty} x^* \quad \forall x_0 \in U$

4 Basin is the whole space

$U = X$

equilibrium

saddle,
unstable node;
 $x' = x$ at 0

stable

center $\ddot{x} + x = 0$: stable, *not*
asympt. stable

Milnor attractor

$x' = -x^2$: basin $[0, \infty)$ yet
unstable — row 1 skipped

asympt. stable

stable node /
focus

globally asympt. stable

$x' = -x$ on $\mathbb{R}$

Lyapunov function

on $U \ni x^*$

- (D1) L continuous
- (D2) $L(x^*) = 0, L(x) > 0$ for $x \neq x^*$
- (D3) nonincreasing along orbits: $t_0 < t_1 \Rightarrow L(\varphi_{t_0}(x_0)) \geq L(\varphi_{t_1}(x_0))$ (*strict:* never =)

differentiable case: $\frac{d}{dt}L(\varphi_t(x)) = \nabla L \cdot f \leq 0$ (Lie derivative)

Lyapunov criterion & LaSalle

$\exists$ Lyapunov function $\Rightarrow$ **stable** (sublevel sets S_ϵ are positively invariant)
 L not constant on any nontrivial orbit $\Rightarrow$ **asymptotically stable** (L is constant on $\omega(x)$, so $\omega(x) = \{x^*\}$)

ω -/ α -limit sets

$\omega(x) = \{y : \exists t_j \rightarrow +\infty, \varphi_{t_j}(x) \rightarrow y\}, \alpha(x): t_j \rightarrow -\infty$

First integral

constant of motion
 $\nabla g \cdot f \equiv 0$. Hamiltonian: $x' = \partial_y H, y' = -\partial_x H$ has $H = \frac{1}{2}|y|^2 + V(x)$ as first integral (not a Lyapunov function in general).
harmonic oscillator: $L = x^2 + y^2$ is both

Linear(ized) stability

$x' = Ax$: all $\text{Re } \lambda_j < 0 \Rightarrow$ asympt. stable; some $\text{Re } \lambda_j > 0 \Rightarrow$ unstable.
maps $x_{k+1} = Ax_k$: multipliers $|\mu_j| < 1 \Rightarrow$ asympt. stable.
 $\text{Re } \lambda_j = 0$ cases undecided by linearization $\rightarrow$ center manifold

Worked example: 1D bifurcation analysis

exam pattern

$x' = f(x, p) = px + 2x^2 - x^3 = x(p + 2x - x^2)$

- steady states:** $x = 0$ and $x_\pm = 1 \pm \sqrt{1+p}$ (exist for $p \geq -1$)
- linear stability:** $\partial_x f = p + 4x - 3x^2$; at $x = 0$: $\lambda = p$ — stable $p < 0$, unstable $p > 0$
- fold** at $(x, p) = (1, -1)$: $f = 0, \partial_x f = 0, \partial_{xx} f = -2 \neq 0, \partial_p f = 1 \neq 0$ ✓
- transcritical** at $(0, 0)$: x_- crosses 0, stability is exchanged
- sketch: branches $x = 0$ and the parabola $p = x^2 - 2x$ in the (p, x) -plane, solid = stable, dashed = unstable



node $b_j = 0, a_1 a_2 > 0$



saddle $b_j = 0, a_1 a_2 < 0$



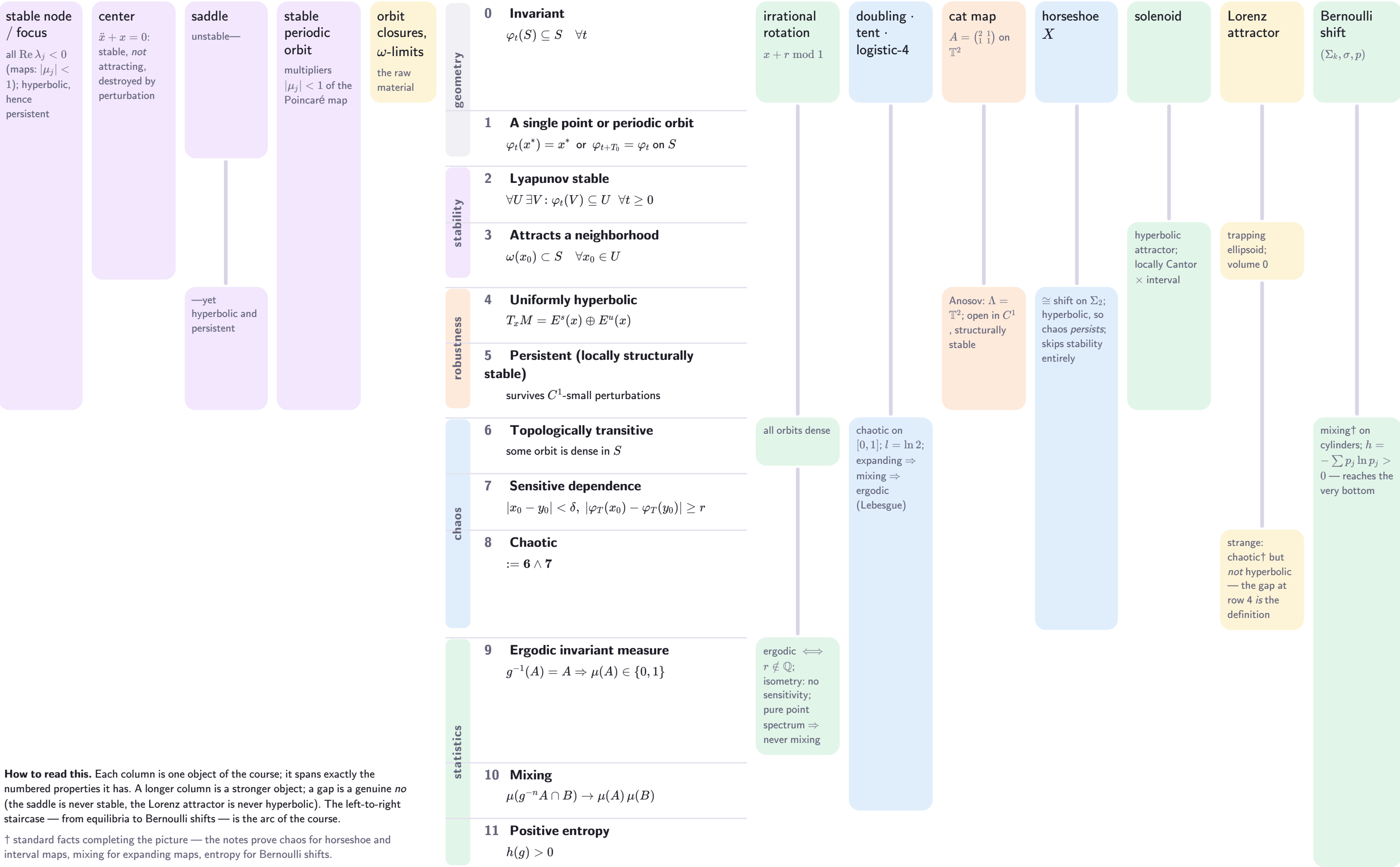
focus $b_1 \neq 0, a_1 \neq 0$



center $a_j = 0, b_1 \neq 0$

$\lambda_{1,2} = a_{1,2} + ib_{1,2}$ eigenvalues of $Df(x^*)$; qualifiers: stable/unstable by sign a_j .

The dynamical hierarchy an invariant object S of a dynamical system (X, T, φ_t) — each column spans exactly the properties established for it; a gap is a genuine *no* (or beyond the notes)



Hyperbolicity & invariant manifolds

§3, §10



flows: x^* hyperbolic $\iff Df(x^*)$ has no eigenvalue with $\operatorname{Re} \lambda = 0$



maps: x^* hyperbolic $\iff$ all multipliers $|\mu| \neq 1$ of $Dg(x^*)$

Manifolds in 60 seconds §3

d -dim **topological manifold**: locally homeomorphic to $\mathbb{R}^d$ (second-countable, Hausdorff); C^k -**manifold**: atlas with C^k chart transitions $h_\alpha \circ h_\beta^{-1}$. **Tangent space** $T_x M$: velocities $\left.\frac{d}{dt}f(\gamma(t))\right|_{t=0}$ of curves through x .

every $d \leq 3$ topological manifold is smoothable — false for $d \geq 4$; examples: $\mathbb{R}^d$, $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, S^d

Invariant splitting

$E^s = \{y : e^{tA}y \rightarrow 0, t \rightarrow +\infty\}$, E^u : $t \rightarrow -\infty$; $\mathbb{R}^d = E^s \oplus E^u$ with projections P^s, P^u .

Stable/unstable manifold theorem flows & diffeos, $f \in C^k$

$$W^s_{\text{loc}}(x^*) = \{y \in U : \varphi_t(y) \in U \ \forall t \geq 0, \ \varphi_t(y) \rightarrow x^*\}$$

$W^{s/u}_{\text{loc}}$ are C^k manifolds, **tangent to $E^{s/u}$ at x^*** ; globalize by $W^{s/u} = \bigcup_t \varphi_t(W^{s/u}_{\text{loc}})$.

constructions: Lyapunov–Perron (variation of constants + contraction) or Hadamard graph transform; maps: $W^s_{\text{loc}} = \{y : g^k(y) \rightarrow x^*\}$

ex. Duffing $\ddot{x} = x - px^3$: saddle at 0, $\dim E^s = \dim E^u = 1$; Arnold cat map on $\mathbb{T}^2$: every point hyperbolic

Hartman–Grobman

x^* hyperbolic $\Rightarrow \varphi_t$ is **locally topologically conjugate** to the linearization e^{tA} , $A = Df(x^*)$. Maps: $h \circ A = g \circ h$, $h = \text{Id} + H$ unique via Banach fixed point.

$\Rightarrow$ near hyperbolic points, the linear picture is the whole local picture (up to homeomorphism — see equivalence ladder)

Non-hyperbolic? — center manifolds

Spectral split

d_c eigenvalues $\operatorname{Re} \lambda = 0$, d_s with $\operatorname{Re} \lambda < 0$, d_u with $\operatorname{Re} \lambda > 0$; E^c = span of critical eigenvectors.

Center manifold theorem $f \in C^k$

$\exists$ local C^k , d_c -dim invariant manifold $W^c_{\text{loc}}(0)$ tangent to E^c ; solutions staying near 0 for $t \rightarrow \pm\infty$ lie on it.

caveats: not unique, possibly not analytic/not C^∞ — e.g. $x' = x^2$, $y' = -y$ has the family $y = p e^{1/x}$ ($x < 0$)

Reduction theorem

With blocks $x' = Bx + F(x, y)$, $y' = Cy + G(x, y)$ ($x \in \mathbb{R}^{d_c}$): locally topologically equivalent to

$$x' = Bx + F(x, H(x)), \quad y' = Cy$$

where $W^c_{\text{loc}} = \text{graph } H$, $H = \mathcal{O}(x^2)$.

Recipe: computing W^c exam workhorse

- shift equilibrium to 0, split into (B, C) blocks (diagonalize)
 - ansatz $y = h(x) = a_2x^2 + a_3x^3 + \dots$
 - invariance equation $\mathcal{N}(h) = Dh(x)[Bx + F(x, h)] - Ch - G(x, h) = 0$
 - match coefficients; approximation thm: $\mathcal{N}(h) = \mathcal{O}(|x|^p) \Rightarrow H = h + \mathcal{O}(|x|^p)$
- ex. $x' = xy$, $y' = -y + px^2$: $a_2 = p$, $W^c : y = px^2 + \mathcal{O}(x^4)$, reduced flow $x' = px^3 + \mathcal{O}(x^5)$.

The equivalence ladder

§4

1 h homeomorphism mapping orbits to orbits

phase portraits agree as sets of curves

2 preserves direction of time

3 preserves time parametrization

$$h \circ \varphi_t = \psi_t \circ h \quad \forall t$$

$$\text{maps: } f = h^{-1} \circ g \circ h$$

4 $h, h^{-1} \in C^k$ (diffeomorphism)

$$\text{flows: } f(x) = (Dh(x))^{-1}g(h(x))$$

topological equivalence

preserves: equilibria, periodic orbits, stability

topological conjugacy

+ periods, entropy
Hartman–Grobman lands here (locally)

C^k -equivalence

C^k -conjugacy

+ eigenvalues / multipliers of linearization

rows 1–2 + 4

Conjugacy in action

tent map $\cong$ logistic map $4y(1 - y)$ via $h(x) = \frac{1 - \cos \pi x}{2}$: $f \circ h = h \circ g$.

to transport a system: $y = h(x) \Rightarrow y' = Dh(h^{-1}(y)) f(h^{-1}(y))$

Flows $\leftrightarrow$ maps

§5

continuous time φ_t

discrete time g^k

equilibrium $f(x^*) = 0$

fixed point $g(x^*) = x^*$

asympt. stable $\iff \operatorname{Re} \lambda_j < 0$

$\iff |\mu_j| < 1$

periodic orbit Γ , period T

fixed point of Poincaré map P

time- T map φ_T

any diffeo, via suspension flow

Poincaré map at $y \in \Gamma$

section $\Sigma = \{x : (x - y) \cdot f(y) = 0\}$ transverse to the flow. Implicit function theorem $\Rightarrow$ return time $\tau \in C^1$, $\tau(y) = T$, and

$$P(x) = \varphi_{\tau(x)}(x) \in C^1(\Sigma \cap B(y; \delta))$$

Multipliers & monodromy

multipliers $\mu_1, \dots, \mu_{d-1}$ = eigenvalues of $DP(0)$ — independent of the point on Γ , the section, and coordinates (local Poincaré maps are conjugate).

variational equation $u' = Df(\Gamma(t))u$; monodromy matrix $M(T)$ has eigenvalues $\{1, \mu_1, \dots, \mu_{d-1}\}$.

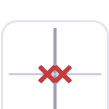
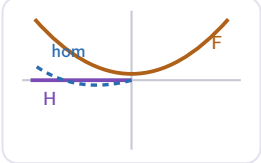
all $|\mu_j| < 1 \Rightarrow \Gamma$ asymptotically stable.

Suspension & obstruction

$g(x) = -\frac{1}{2}x$ is *not* a time- T map of any flow on $\mathbb{R}$ (orientation). But every diffeo g of $\mathbb{R}^d$ is the return map of a flow on the $(d+1)$ -dim mapping torus $M = (\mathbb{R}^d \times [0, T])/\sim$, $(g(x), 0) \sim (x, T)$.

Bifurcation loss of equivalence under parameter variation

Families $x' = f(x, p)$, $y' = g(y, q)$ are **fiber topologically equivalent** if a homeomorphism $\xi(p) = q$ matches parameters and each $f(\cdot, p) \sim g(\cdot, \xi(p))$ topologically. A **bifurcation diagram** stratifies parameter space by equivalence class; an **unfolding** contains the bifurcation persistently; the minimal number of parameters is the **codimension**.

λ / μ	bifurcation	genericity conditions	topological normal form	what happens
	fold (saddle-node) flows · codim 1	$f(0, 0) = 0, \partial_x f(0, 0) = 0;$ $\partial_{xx} f(0, 0) \neq 0, \partial_p f(0, 0) \neq 0$	$y' = q \pm y^2$	two equilibria collide and annihilate $x' = p - x^2$: $\pm\sqrt{p}$ for $p > 0$, none for $p < 0$
	transcritical flows · constrained	$\lambda = 0$ with fixed line of equilibria $x = 0$	$y' = y(q - y)$	two equilibria cross and exchange stability
	Hopf flows, $d \geq 2$ · codim 1	$\lambda_{1,2}(p) = \mu(p) \pm i\omega(p)$, $\mu(0) = 0, \omega(0) > 0$; first Lyapunov coeff. $l_1 \neq 0$	$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} q & -1 \\ 1 & q \end{pmatrix} y \pm (y_1^2 + y_2^2) y$	limit cycle born; $l_1 < 0$ supercritical (stable cycle), $l_1 > 0$ subcritical
	fold of maps maps · codim 1	$\mu = 1$; $\partial_{xx} g \neq 0, \partial_p g \neq 0$	$y \mapsto q + y \pm y^2$	fixed points collide and annihilate
	flip (period doubling) maps · codim 1	$\mu = -1$; non-degeneracy + transversality	$y \mapsto -(1 + q) y \pm y^3$	fixed point loses stability, stable period-2 orbit emerges; cascades $\rightarrow$ Feigenbaum
	Neimark–Sacker maps · codim 1	complex pair of multipliers crosses the unit circle	on W^c : rotation + cubic terms; closed invariant curve of radius $\mathcal{O}(\sqrt{q})$	closed invariant curve bifurcates Hénon map $(x_1, x_2) \mapsto (x_2, p_1 - p_2 x_1 - x_2^2)$: NS, flip & fold regions
	Bogdanov–Takens flows · codim 2	double zero eigenvalue, nilpotent $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$	$y_1' = y_2,$ $y_2' = q_1 + q_2 y_1 + y_1^2 \pm y_1 y_2$	 fold $4q_1 = q_2^2$, Hopf $q_1 = 0, q_2 < 0$, homoclinic curve between — rescaling $\rightarrow$ Hamiltonian $\frac{1}{2}Y_2^2 - \frac{1}{2}Y_1^2 + \frac{1}{3}Y_1^3$ with homoclinic loop
	homoclinic loop Andronov–Leontovich $\mathbb{R}^2$ · global, codim 1	homoclinic orbit to saddle, $\lambda_1 < 0 < \lambda_2$; saddle quantity $\sigma = \lambda_1 + \lambda_2 \neq 0$; split function $\xi'(0) \neq 0$	$\xi(p)$ = signed distance of W^u, W^s on a section Σ near the saddle	unique limit cycle bifurcates from the loop; $\sigma < 0$: stable, $\sigma > 0$: unstable
	Shilnikov $\mathbb{R}^3$ · global	homoclinic orbit to saddle-focus: $\lambda_1 > 0, \lambda_{2,3} = \alpha \pm i\omega, \alpha < 0$; $\sigma = \lambda_1 + \alpha > 0, \lambda_2 \neq \lambda_3$	spiral return map intersects itself transversally	infinitely many saddle periodic orbits accumulate on the loop $\Rightarrow$ chaos ex. FitzHugh–Nagumo traveling waves, $0 < \varepsilon \ll 1$

Homoclinic orbit: $\gamma(t) \rightarrow x^*$ for $t \rightarrow \pm\infty$; heteroclinic: different limits. Structurally unstable in $\mathbb{R}^2$ (Peixoto) but persistent phenomena occur for $d \geq 3$.

Structural stability

§6

Definition

$f \in C^1(M)$ is **structurally stable** if $\exists \varepsilon > 0$: every g with $\|f - g\|_{C^1} < \varepsilon$ generates a flow topologically equivalent to φ_t^f .
 $\|f\|_{C^k} = \sum_{|\alpha| \leq k} \sup_K |D^\alpha f|$ on compact K — a Banach space

Nonwandering set

$x \in \text{NW}(\varphi)$: every neighborhood $U \ni x, \forall t_0 \exists t > t_0: \varphi_t(U) \cap U \neq \emptyset$.

Peixoto compact 2-manifold

$f \in C^1$ structurally stable $\iff$

- (T1) finitely many equilibria & periodic orbits, all hyperbolic
- (T2) no saddle–saddle connections: $W^s(x) \cap W^u(y) = \emptyset$
- (T3) $\text{NW}(\varphi) = \{\text{equilibria}\} \cup \{\text{periodic orbits}\}$

M orientable $\Rightarrow$ structurally stable fields are **open and dense** in $C^1(M)$. Fails for $d \geq 3$.

ex. torus flow $x' = \omega_1, y' = \omega_2$: structurally *unstable* ($\omega_1/\omega_2 \in \mathbb{Q}$: all orbits periodic; irrational: all orbits dense)

Morse–Smale system

(D1) finitely many equilibria + periodic orbits, all hyperbolic; (D2) all $W^s \cap W^u$ transversal; (D3) $\text{NW} = \text{equilibria} \cup \text{periodic orbits}$.

Thm: Morse–Smale flows on compact M are structurally stable (any dim) — but *not generic* for $d \geq 3$.

Generic property

holds on a **residual** set = countable intersection of open dense sets.

ex. hyperbolicity is generic in $\mathbb{R}^{d \times d}$

Hyperbolic sets

§7

Hyperbolic set $g \in C^1$ diffeo, Λ compact invariant

$\exists C > 0, \lambda \in (0, 1)$ and subspaces $E^s(x), E^u(x) \subset T_x M$:

- (D1) invariance: $Dg_x E^{s/u}(x) = E^{s/u}(g(x))$
- (D2) $|Dg_x^j v| \leq C \lambda^j |v|$ on $E^s, j \geq 0$
- (D3) $|Dg_x^{-j} v| \leq C \lambda^j |v|$ on E^u
- (D4) splitting: $T_x M = E^s(x) \oplus E^u(x)$

Anosov diffeomorphism: $\Lambda = M$. ex. cat map $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ on $\mathbb{T}^2$.

Shadowing

δ -pseudo-orbit: $d(g(x_k), x_{k+1}) \leq \delta$. Near a hyperbolic Λ : $\forall \varepsilon \exists \delta$ such that every δ -pseudo-orbit is ε -**shadowed** by a true orbit.
numerics with rounding errors still trace real orbits

Persistence

C^1 -small perturbations f of g : nearby compact invariant sets remain hyperbolic, and $\exists$ hyperbolic K with $\psi \circ f|_K = g|_\Lambda \circ \psi, \psi$ homeo near Id.

Cor: Anosov diffeos are open in C^1 and structurally stable.

the horseshoe’s invariant set is hyperbolic $\Rightarrow$ chaos persists $\rightarrow$ see the road to chaos

The road to chaos

§14, §15

Periodically perturbed Hamiltonian system

$x' = f(x) + \varepsilon g(x, t), f = (\partial_{x_2} H, -\partial_{x_1} H)^\top, g$ T -periodic; unperturbed homoclinic orbit $\gamma(t)$ to a saddle p . Autonomize on $\mathbb{R}^2 \times S^1$, take the stroboscopic Poincaré map $P_\varepsilon^{t_0}$ — the saddle persists: $p_\varepsilon^{t_0} = p + \mathcal{O}(\varepsilon)$.

$\downarrow$ leading-order splitting of W^s, W^u

Melnikov function

$$M(t_0) = \int_{-\infty}^{\infty} f(\gamma(t - t_0)) \wedge g(\gamma(t - t_0), t) dt$$

signed distance: $d(t_0) = \varepsilon \frac{M(t_0)}{|f(\gamma(0))|} + \mathcal{O}(\varepsilon^2), a \wedge b = a_1 b_2 - a_2 b_1$.

simple zero of $M \Rightarrow W^s(p_\varepsilon) \cap W^u(p_\varepsilon)$; no zeros $\Rightarrow$ no intersection.

Duffing $x'' = x - x^3 + \varepsilon(\alpha \cos t - x')$, $\gamma_\pm = \pm(\sqrt{2} \operatorname{sech} t, -\sqrt{2} \operatorname{sech} t \tanh t)$: $M(t_0) = -\frac{4}{3} + \sqrt{2} \alpha \pi \operatorname{sech}(\frac{\pi}{2}) \sin t_0$ — simple zeros for α large enough

$\downarrow$ transverse homoclinic point $q \in W^s(0) \cap W^u(0)$

Smale–Birkhoff homoclinic theorem

diffeo with hyperbolic saddle + transverse homoclinic point $\Rightarrow$ some iterate g^κ has an invariant set X conjugate to a **horseshoe**.

λ -lemma: a disk D transverse to W^s has iterates $g^n(D)$ accumulating C^1 on W^u — this folds a rectangle near q into a horseshoe.

(Transverse homoclinic points occur only in maps / Poincaré maps, never in planar flows.)

$\downarrow$ stretch $\mu > 2$, contract $\lambda < \frac{1}{2}$, fold

Smale horseshoe $\cong$ full shift

$X = \{x : g^k(x) \in Y = [0, 1]^2 \forall k \in \mathbb{Z}\}$; itinerary map $\varphi(x)_j = a_j$ with $g^j(x) \in H_{a_j}$ is a homeomorphism conjugating $g|_X$ to the shift σ on $\Sigma_2 = \{0, 1\}^\mathbb{Z}$.

generalized horseshoe: N strips + cone conditions $\Rightarrow$ shift on N symbols

$\downarrow$ count orbits symbolically

Consequences

- countably many (saddle) periodic orbits, of every period
- uncountably many aperiodic orbits; a dense orbit
- X is a hyperbolic set $\Rightarrow$ the whole picture **persists** under perturbation
- $g|_X$ is chaotic; Lyapunov exponents $\ln \mu > 0 > \ln \lambda$
alternative entrance: Shilnikov saddle-focus loop (zoo) also forces horseshoes

Symbolic dynamics: the universal model

§1, §13, §22

Shift spaces

Σ_k = bi-infinite sequences over $\{0, \dots, k-1\}$; shift $(\sigma\omega)_j = \omega_{j+1}$. Periodic orbits = periodic sequences; a sequence containing every finite word has a dense orbit.

One model, many systems

- horseshoe $g|_X \cong \sigma$ on Σ_2 (itineraries through H_0, H_1)
- tent map $p=2$: semi-conjugate to Σ_2^+ via binary coding $H(s) = \sum_j s_j 2^{-(j+1)}$
- Bernoulli measure p on $\Sigma_k \rightarrow$ the probabilistic model; entropy $-\sum p_j \ln p_j$

prove chaos once for σ , transport it along conjugacies

Normal forms: the reduction pipeline

§9, §10, §11

non-hyperbolic equilibrium

$d_c > 0$ critical eigenvalues block B ; $A = Df(0)$



center-manifold reduction

restrict to W_{loc}^c : dynamics in $\mathbb{R}^{d_c}$ decide everything



kill non-resonant terms

$x = y + h^{(k)}(y)$, solve $Lh^{(k)} = f^{(k)}$ order by order



resonant normal form

$y' = Ay + r^{(2)} + \cdots + r^{(k)} + \mathcal{O}(|y|^{k+1})$



truncate + unfold

add parameters $q \rightarrow$ topological normal forms of the zoo

Homological equation

$L : H_k \rightarrow H_k$ on homogeneous vector polynomials, $Lh = Dh \cdot Ay - A h(y) = [h, Ay]$.

Solvable part of $f^{(k)}$ is removed; **resonant terms** span a complement $R_k = \ker L^*$ of range L (if no generalized 0-eigenvectors: $H_k = \ker L \oplus \text{range } L$).

in practice: write L as matrix $M^{(k)}$ in monomial basis, e.g. $k = 2$: $\{x^2, xy, y^2\} \otimes \{\partial_x, \partial_y\} \cong \mathbb{R}^6$; resonant terms from $\ker(M^{(k)})^\top$

Hopf normal form eigenvalues $\pm i\omega$

$$y' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} y + (y_1^2 + y_2^2) \begin{pmatrix} p_1 & -p_2 \\ p_2 & p_1 \end{pmatrix} y + \mathcal{O}(|y|^4)$$

cubic terms are resonant — they cannot be removed; $p_1 = l_1$ decides super/subcritical

Jets & determinacy §11

k -jet $J^k F =$ Taylor polynomial to order k . BT 2-jet $y'_1 = y_2, y'_2 = y_1^2 \pm y_1 y_2$ is **topologically determinate**: blow-up $\Phi(\theta, r) = (r \cos \theta, r \sin \theta)$, desingularize $\bar{F} = r^{-l}(D\Phi^{-1} \circ F \circ \Phi)$ — the origin blows up to a circle of hyperbolic equilibria, so Hartman–Grobman finishes the classification.

One-dimensional maps

§16

1 ◀ 2 ◀ 4 ◀ 8 ◀ ⋯ ◀ 2ⁿ·7 ◀ 2ⁿ·5 ◀ 2ⁿ·3 ◀ ⋯ ◀ 9 ◀ 7 ◀ 5 ◀ 3

Sharkovskii $g \in C^0([a, b], \mathbb{R})$

periodic point of minimal period $n \Rightarrow$ periodic orbits of every period $k \triangleleft n$.

Li–Yorke: period three $\Rightarrow$ all periods (“period three implies chaos”).

tent map $p = 2$: $\frac{2}{7} \mapsto \frac{4}{7} \mapsto \frac{6}{7} \mapsto \frac{2}{7}$ is a 3-orbit

Schwarzian derivative

$$Sg = \frac{g'''}{g'} - \frac{3}{2} \left(\frac{g''}{g'} \right)^2$$

minimum principle: $Sg < 0 \Rightarrow |g'|$ has no interior local minimum.

unimodal: $g(0) = g(1) = 0$, unique max; S-unimodal: also $Sg < 0$

Singer

$g \in C^3(I, I)$, $Sg < 0$, n critical points $\Rightarrow$ at most $n + 2$ attracting periodic orbits.

logistic $px(1 - x)$: $Sg = \frac{-6}{(1-2x)^2} < 0$, one critical point $\Rightarrow \leq 3$

Renormalization & universality

§17

Idea

act on maps by *rescaling* phase space, time (iterate), and parameters; self-similar dynamics = fixed point of the renormalization operator.

fold: $g^2(y; q) \approx 2q + y - 2y^2$ rescales back to a fold ($\alpha = 2, \beta = 4$) $\Rightarrow$ passage time $N(q) \sim q^{-1/2}$ (critical slowing down)

Feigenbaum universality S-unimodal maps

flip cascade $p_0 < p_1 < \cdots \rightarrow p_\infty$ (logistic: $p_0 = 3$, 2-cycle multiplier $4 + 2p - p^2$),

$$\delta_F = \lim_n \frac{p_n - p_{n+1}}{p_{n+1} - p_n} \approx 4.6692, \quad d_n = \kappa_2 \alpha_F^{-n}, \quad \alpha_F \approx -2.5029$$

d_n : distance of the superstable 2^n -cycle to the critical point $x = \frac{1}{2}$ (superstable: all multipliers = 0, i.e. $\frac{1}{2}$ in the cycle)

fixed-point equation $g(x) = \alpha g^2(x/\alpha)$, $g(0) = 1, g'(0) = 0$; $\delta_F =$ the unique unstable eigenvalue of the linearized operator — the same constants for *every* such family.

0 Trapping region

U compact, $\varphi_t(U) \subset \text{int}(U) \ \forall t > 0$

1 Attracting set

$A = \bigcap_{t \geq 0} \varphi_t(U)$
closed, invariant; $\omega(x_0) \subset A$ on U ; hyperbolic
 $x^* \in A \Rightarrow W^u(x^*) \subset A$

2 Indecomposable

no smaller attracting set inside

3 Chaotic on A

sensitive + transitive

4 Hyperbolic set

uniform splitting (D1)–(D4)

attracting set

$r' = 1 - r$ on circle:
{ $r=1$ } attracts,
but most orbits go
to one point

attractor

stable equilibria,
stable limit cycles

chaotic /
strange
attractor

strange = chaotic
 $\wedge$ not hyperbolic:
Lorenz

hyperbolic
attractor

Smale–Williams
solenoid: locally
Cantor $\times$ interval

Lorenz system $\sigma = 10, \rho = 28, \beta = 8/3$

$x'_1 = \sigma(x_2 - x_1), x'_2 = \rho x_1 - x_2 - x_1 x_3, x'_3 = x_1 x_2 - \beta x_3$; the ellipsoid $\sigma x_1^2 + x_2^2 + \beta(x_3 - \frac{\rho + \sigma}{2})^2 \leq c$ is trapping. $\nabla \cdot f = -(\sigma + 1 + \beta) < 0 \Rightarrow$ the attractor has **volume zero**.

Smale–Williams solenoid hyperbolic attractor

$g(\theta, x, y) = (2\theta, \lambda x + \frac{1}{2} \cos 2\pi\theta, \lambda y + \frac{1}{2} \sin 2\pi\theta)$ on $S^1 \times D^2, \lambda \in (0, \frac{1}{2})$; $A = \bigcap_k g^k(S^1 \times D^2)$.

Milnor attractor measure-theoretic

closed invariant A with $\mu(B(A)) > 0$ and no proper closed $\tilde{A} \subset A$ with $\mu(B(A) \setminus B(\tilde{A})) = 0$.

ex. $x' = -x^2$: $x^* = 0$ attracts $[0, \infty)$ although unstable

Fractal dimension

box-counting: $\dim_B(S) = \lim_{r \rightarrow 0} \frac{\ln N(r, S)}{\ln(1/r)}$; middle-third Cantor set: $\frac{\ln 2}{\ln 3} \approx 0.63$. Strange attractors typically have non-integer dimension.

Milnor attractor (positive-measure basin) — see stability ladder, row 2

1 Sensitive dependence

$\exists r > 0 \ \forall \delta \ \exists y_0, T :$
 $|x_0 - y_0| < \delta, \ |\varphi_T(x_0) - \varphi_T(y_0)| \geq r$

2 Topological transitivity

some forward orbit is dense in X

3 Chaotic := 1 $\wedge$ 2

on an invariant set S

4 Positive Lyapunov exponent

$l_j(x_0) = \lim_k \frac{1}{k} \ln \lambda_j^{(k)}$
 $\lambda_j^{(k)}$ singular values of $Dg_{x_0}^k$; Liouville:
 $\sum_j l_j = \ln |\det Dg|$ (if constant)

doubling

$2x \bmod 1$;
 $l = \ln 2$

tent $\cong$
logistic

$p=2$ / $p=4$:
conjugate!
 $l = \ln 2$

shift σ

on Σ_2 ;
universal
model; row 4
n/a

horseshoe

$l_1 = \ln \mu,$
 $l_2 = \ln \lambda$

rotation

irrational:
dense orbits,
no sensitivity

0 Fixed: $g(x) = x$

Fix(g)

Per(g)

NW(g)

CR(g)

chain classes =
equivalence
classes of two-
way δ -chains

1 Periodic: $g^n(x) = x$

2 Nonwandering

$\forall U \ni x$, some $\varphi_t(U) \cap U \neq \emptyset, t$ arbitrarily large

3 Chain recurrent

$\forall \delta > 0$ a δ -pseudo-orbit loops $x \rightarrow x$

Conley homeo of compact metric X

$\exists$ **complete Lyapunov function** $L \in C(X, \mathbb{R})$:

- (T1) $L(g(x)) < L(x)$ off CR(g)
- (T2) on CR(g): $L(x) = L(y) \iff x, y$ chain equivalent
- (T3) $L(\text{CR}(g))$ compact, nowhere dense (Cantor-like)

moral: every system = gradient-like part + chain-recurrent part.

Construction via attractor–repellor pairs

dual repellor $A^* = \bigcap_{k \geq 0} g^{-k}(X \setminus \overline{U})$ of an attractor A with isolating U ; there is $L_i \in C(X, [0, 1])$ with $L_i|_{A_i} = 0, L_i|_{A_i^*} = 1$, strictly decreasing elsewhere. Then

$$\text{CR}(g) = \bigcap_i (A_i \cup A_i^*), \quad L = \sum_i \frac{2}{3^i} L_i$$

$L(\text{CR})$ lands in the middle-third Cantor set — hence nowhere dense

0 Invariant (probability) measure

$\mu(g^{-1}(A)) = \mu(A) \quad \forall A \in \mathcal{F}$
 $\Rightarrow \int f \, d\mu = \int f \circ g \, d\mu$

rotation
 $r \in \mathbb{Q}$
Lebesgue
invariant; not
ergodic

rotation
 $r \notin \mathbb{Q}$
ergodic (Fourier)
but **not** weak
mixing: pure
point spectrum
 $e^{2\pi i n r}$

expanding
maps
 $\alpha x \bmod 1, \alpha \in \mathbb{Z}$
, $|\alpha| > 1$; dyadic
intervals refine
uniformly

Bernoulli shift
 (Σ_k, σ, p) ;
entropy
 $-\sum p_j \ln p_j$

1 Ergodic

$g^{-1}(A) = A \Rightarrow \mu(A) \in \{0, 1\}$
 $\iff$ invariant $f \in L^2$ const a.e. $\iff$ time avg =
space avg a.e. $\iff \frac{1}{n} \sum_j \mu(g^{-j} A \cap B) \rightarrow$
 $\mu(A)\mu(B)$

2 Weak mixing

Cesàro-averaged correlations vanish:
 $\frac{1}{n} \sum_j |\mu(g^{-j} A \cap B) - \mu(A)\mu(B)| \rightarrow 0$

3 Strong mixing

$\mu(g^{-n} A \cap B) \rightarrow \mu(A) \mu(B)$
correlations decay: the system forgets its initial state

strong mixing $\Rightarrow$ weak mixing $\Rightarrow$ ergodic; all implications strict.

Ergodic theorems

Krylov–Bogolyubov

X compact metric, g continuous $\Rightarrow$ an invariant Borel probability measure exists (empirical averages + Riesz representation)

Gauss map invariant measures can be exotic

$g(x) = \frac{1}{x} - \lfloor \frac{1}{x} \rfloor$ on $[0, 1)$ preserves $\mu((a, b)) = \frac{1}{\ln 2} \int_a^b \frac{dy}{1+y}$ — the shift on continued-fraction digits.

Poincaré recurrence

g measure-preserving, $\mu(B) > 0 \Rightarrow$ a.e. $x \in B$ returns to B infinitely often.

Birkhoff $f \in L^1$

$\frac{1}{n} \sum_{j=0}^{n-1} f(g^j(x)) \longrightarrow f^*(x)$ a.e.
 f^* invariant, $\int f^* d\mu = \int f d\mu$; **ergodic case:** $f^* = \int f d\mu$ a.e. — time average = space average.
von Neumann: also $S_n f \rightarrow f^*$ in L^p ; key lemma: maximal ergodic theorem $\int_{\{F_N > 0\}} f \, d\mu \geq 0$

Koopman & Perron–Frobenius operators

$L_g f = f \circ g$ on L^p (isometry); adjoint P_g : $\langle P f, h \rangle = \langle f, L h \rangle$.
• ergodic $\iff 1$ is a simple eigenvalue of L_g
• invertible + continuous spectrum (1 the only eigenvalue) $\Rightarrow$ weak mixing
• strong mixing $\iff \langle P^n f, h \rangle \rightarrow \langle f, 1 \rangle \langle 1, h \rangle$
irrational rotation: eigenfunctions $e^{2\pi i n x}$ span L^2 (pure point) $\Rightarrow$ not weak mixing

Entropy of a partition $\rightarrow$ of a map

$H(\mathcal{A}) = -\sum_j \mu(A_j) \ln \mu(A_j)$
 $h(g, \mathcal{A}) = \lim_n \frac{1}{n} H\left(\bigvee_{j=0}^{n-1} g^{-j} \mathcal{A}\right), \quad h(g) = \sup_{\mathcal{A}} h(g, \mathcal{A})$
 $\mathcal{A} \vee \mathcal{B} = \{A_k \cap B_n\}$: joint refinement; $H(\mathcal{A} \vee \mathcal{B}) = H(\mathcal{A}) + H(\mathcal{B} \mid \mathcal{A})$

Kolmogorov–Sinai & invariance

h is a **conjugacy invariant**; for a generating partition $\mathcal{A}$: $h(g) = h(g, \mathcal{A})$.

Conditional entropy & rules

$H(\mathcal{A} \mid \mathcal{B}) = -\sum_{i,j} \mu(B_j) \mu(A_i \mid B_j) \ln \mu(A_i \mid B_j)$
• $H(\mathcal{A} \vee \mathcal{B}) = H(\mathcal{A}) + H(\mathcal{B} \mid \mathcal{A})$; $\mathcal{A} \subseteq \mathcal{B} \Rightarrow H(\mathcal{A}) \leq H(\mathcal{B})$
• $h(g, \mathcal{A}) \leq H(\mathcal{A})$; $h(g, \mathcal{A}) \leq h(g, \mathcal{B}) + H(\mathcal{A} \mid \mathcal{B})$
• g -invariance: $H(g^{-1} \mathcal{A} \mid g^{-1} \mathcal{B}) = H(\mathcal{A} \mid \mathcal{B})$

Why entropy is finer than spectrum

Bernoulli shifts: $h = -\sum p_j \ln p_j$; $p = (\frac{1}{2}, \frac{1}{2})$: $h = \ln 2$, $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: $h = \ln 3 \Rightarrow$ **not conjugate**, though spectrally indistinguishable.
entropy $\approx$ average information produced per iteration

How to read this sheet

The lattice panels

Numbered rows are **properties**; each colored column is a **notion** that satisfies exactly the rows it covers. Longer column = stronger notion; a gap in a column means that property is genuinely skipped (e.g. a Milnor attractor need not be Lyapunov stable).

Color code

light purple local theory light orange robustness & bifurcations light blue chaos light green statistics light yellow examples & bridges

Two deviations from the notes deliberate

- Lorenz parameters printed as $\sigma = 10, \rho = 28$ (standard); the notes state $\sigma = 28, \rho = 10$ — a typo given the equations' form.
- Sharkovskii ordering printed in the standard form ending $\cdots \triangleleft 9 \triangleleft 7 \triangleleft 5 \triangleleft 3$; the notes' line reads “ $\cdots \triangleleft 3 \triangleleft 5 \triangleleft \cdots$ ”, inconsistent with Li–Yorke.

The three big bridges

- Hartman–Grobman**: hyperbolic local dynamics = linear dynamics (up to conjugacy)
- Smale–Birkhoff**: one transverse homoclinic point $\Rightarrow$ full symbolic chaos
- Birkhoff ergodic**: time averages = space averages (ergodic case)